

HARSHINEE SRIRAM

Vancouver, BC, Canada

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EDUCATION

The University of British Columbia, Vancouver

2022 – 2027 (If the stars so align)

PhD in Computer Science

Vancouver, BC, Canada

- Advisors: [Margo Seltzer](#) and [Xin Tang](#)
- **Research Focus Areas (including but not limited to):** Virtual Cells, AI for Biology, Drug Discovery, and Diagnosis, Multimodal Learning (Video, Text, Gaze/Eye-Movements), Sparse Diagnostic Models, and Interpretable and Explainable ML.

WORK EXPERIENCE

Amazon

Vancouver, BC, Canada

Applied Scientist II Intern

June 2026 – September 2026

- Will be a part of the Alexa Daily Essentials (ADE) Science & Analytics team.

Applied Scientist II Intern

June 2025 – September 2025

- Researched and developed novel **Human-in-the-Loop** ReAct **Agentic Workflows** for [Alexa+](#) using **MCP tool calling** and **Multimodal Foundational Models** to extend generative AI capabilities beyond static interactions.
- Benchmarked performance of agentic workflows against WebVoyager and Online-Mind2Web using **Amazon Nova Act** and **AWS Strands SDK**, to identify critical bottlenecks in current planning and reasoning paradigms.
- Prototyped large-scale integrations with AWS services to validate the scalability of agent-based multimodal systems.

UBC - Amazon AWS Cloud Innovation Centre (CIC)

Vancouver, BC, Canada

Applied Scientist Intern

September 2021 – April 2025

- **Generative AI Alignment (BC Govt)** [\[Article\]](#) [\[Code\]](#): Architected a **RAG** pipeline on AWS Bedrock with **Guardrails** to enforce safety alignment and mitigate hallucinations while analyzing high-stakes government educational data.
- **Large-Scale LLM Personalization** [\[Article\]](#) [\[Code\]](#): Engineered a history-aware RAG system using **LLaMa 3.1 70B** and **PostgreSQL** vector stores to model student mastery and deliver personalized, instruction-tuned pedagogical support.
- **Entity Resolution for Knowledge Graphs** [\[Article\]](#) [\[Code\]](#): Developed an automated ETL pipeline leveraging **AWS Comprehend NER** and fuzzy matching to construct knowledge graphs from unstructured text, to enable semantic search via **AWS Glue**.
- **Semantic Search & Optimization** [\[Article\]](#) [\[Code\]](#): Fine-tuned a Transformer model on **AWS SageMaker (PyTorch + CUDA)** to evaluate course flexibility; deployed a scalable MLOps workflow that earned a **CUCCIO 2024 Finalist** award.

RESEARCH PUBLICATIONS

1. **Harshinee Sriram**, Jon Donnelly, Alex He-Mo, Margo Seltzer, and Xin Tang. "SWARM-FR: Benchmarking Virtual Cell Metrics." Submitted to NeurIPS 2026 Evaluations and Datasets Track.
2. Vedant Bahel, **Harshinee Sriram**, and Cristina Conati. "Personalizing explanations of AI-driven hints to users cognitive abilities: an empirical evaluation." In the 26th International Conference on Artificial Intelligence in Education (AIED) 2025. [\[Link\]](#) (*Best Paper Nominee*)
3. Shih-Han Chou, Miini Teng, **Harshinee Sriram**, Chuyuan Li, Giuseppe Carenini, Cristina Conati, Thalia S. Field, Hyeju Jang, and Gabriel Murray. "Multimodal Classification of Alzheimer's Disease by Combining Facial and Eye-Tracking Data." In Machine Learning for Health (ML4H), pp. 219-232. PMLR, 2025. [\[Link\]](#) (*Spotlight*)
4. Vedant Bahel, **Harshinee Sriram**, and Cristina Conati. "Initial results on personalizing explanations of AI hints in an ITS." In Proceedings of the 32nd ACM Conference on User Modeling, Adaptation and Personalization UMAP 2024. [\[Link\]](#)
5. Omair Shahzad Bhatti, **Harshinee Sriram**, Abdulrahman Mohamed Selim, Cristina Conati, Michael Barz, and Daniel Sonntag. "Detecting When a User Disagrees with generated Captions." In Companion Proceedings of the 26th International Conference on Multimodal Interaction (ICMI), pp. 195-203. 2024. [\[Link\]](#) (*Oral*)
6. **Harshinee Sriram**, Cristina Conati, and Thalia Field. "Classification of Alzheimer's Disease with Deep Learning on Eye-tracking Data." In International Conference on Multimodal Interaction (ICMI), pp. 104-113. 2023. [\[Link\]](#)
7. Anuj Harisinghani, **Harshinee Sriram**, Cristina Conati, Giuseppe Carenini, Thalia Field, Hyeju Jang, and Gabriel Murray. "Classification of Alzheimer's using Deep-learning Methods on Webcam-based Gaze Data." Proceedings of the ACM on Human-Computer Interaction 7, no. ETRA (2023): 1-17. [\[Link\]](#) (*Oral*)
8. **Harshinee Sriram** and Cristina Conati, "Evaluating the overall sensitivity of saliency-based explanation methods.", International Joint Conference on Artificial Intelligence (IJCAI) Workshop on Explainable Artificial Intelligence 2023. [\[Link\]](#) (*Oral*)